Lesson for the course "Human-Computer Interaction"

Attention and memory

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Our attention is limited, our memory is imperfect

- Human attention and memory have strengths and weaknesses
- Memory:
  - **Short-term memory**
    - Covers situations in which information is retained for intervals ranging from a fraction of a second to a few minutes
  - **Long-term memory**
    - Covers situations in which information is retained over longer periods (hours, days, years, even lifetimes)

Short-term memory

- It is a **combination** of phenomena involving
  - **Perception** (i.e., the inputs from the sensory channels)
  - **Retrieval from long-term memory**
  - **Attention**
    - Mechanisms that focus our awareness on a very small subset of the perceptions and activated long-term memories, and ignore everything else
- **Working memory** is the main part of the short-term memory
  - It is everything we are **conscious** of at any moment
  - Basically, it is **the current focus of attention**
  - It integrates information from all of our sensory modalities and our long-term memory

Attention: is highly focused and selective

- Attentional mechanisms **filter very selectively** the inputs of your sensorial channels
  - You are unaware of most of what is going on around you now
- **Our perception is filtered and biased by our goals**
  - E.g., when searching for a friend in a shopping mall, your visual system primes itself to notice people who look like your friend and barely notice everything else. Simultaneously, your auditory system primes itself to notice voices that sound like your friend's voice

Attention: is highly focused and selective

- Our attention is also captured by
  - **Movement**, especially movement near or toward us
  - **Threats**, anything that signals or portends danger to us or people in our care
  - **Faces of other people**, we are primed from birth to notice faces more than other objects in our environment
  - **Sexual stimuli**
  - **Food**

- These things, along with our current goals, draw our attention **involuntarily**

Characteristics of working memory

- Basically, it is the current focus of attention
- Working memory has **low capacity**
  - "Magic number" 7 ± 2: the limit on the number of **simultaneous unrelated items** in human working memory (items are e.g., words, numbers, images, sounds)
  - However, this number resulted from an experiment with college students (i.e., not the average person)
  - Subsequent experiments with noncollege students revealed that the actual average number is **4 ±1**
- Working memory is **volatile**
  - Focusing attention on new information can easily be lost from working memory
  - Losing items from your working memory corresponds to forgetting or losing track of something you were doing

Implications of working memory characteristics for UI design

- User interfaces should **help people remember essential information** from one moment to the next
- **Don't require people to remember system status or what they have done**, because their attention is focused on their primary goal and progress toward it
- We see some examples:
  - Modes
  - Search results
  - Calls to action
  - Instructions
  - Navigation depth

Modes

- Modes: **when an action on a UI corresponds to a different effect depending on what mode the system is in**
  - E.g., In a car, pressing the accelerator pedal can move the car either forwards, backwards, or not at all, depending on the transmission is in drive, reverse or neutral; the transmission sets a mode in the car's UI
- Advantage: the **same control** provides different functions in different modes
- Disadvantage: people often **forget** what mode the system is in and do the wrong thing by mistake

Modes

- Guideline: **either avoid designs that have modes or provide adequate feedback about which mode the system is in**
  - Working memory is too unreliable for designers to assume that users can, without continuous feedback, keep track of what mode the system is in

Search results

- When people use a search function on a computer to find information, they enter the search terms, start the search, and then review the results
- Evaluating the results often requires knowing what the search terms were
- When the results appear, a person's attention naturally turns away from what he or she entered and toward the results
- Working memory is volatile: people viewing search results often do not remember the search terms they just typed!
- Guideline: **always show the user what s/he searched**

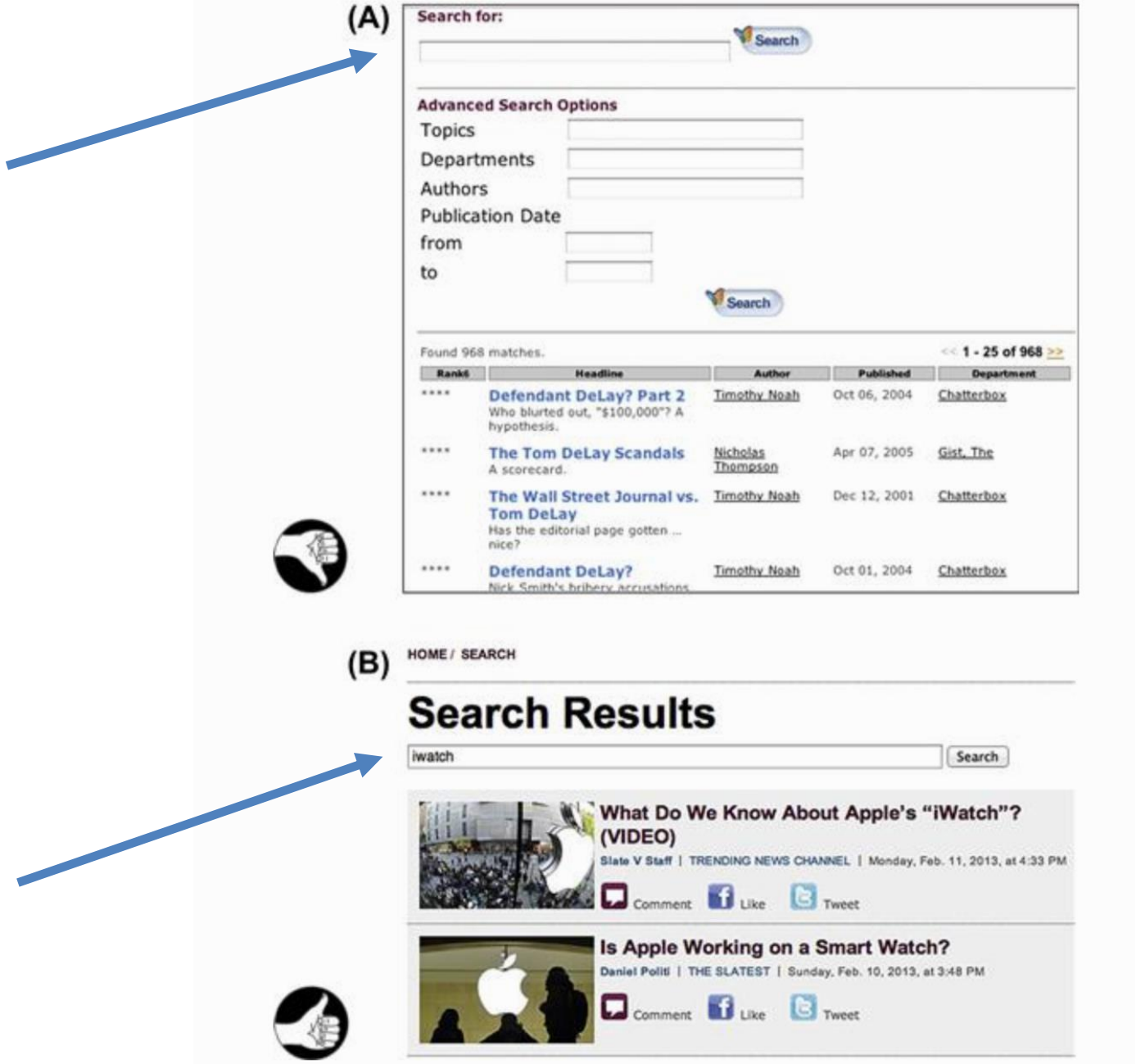
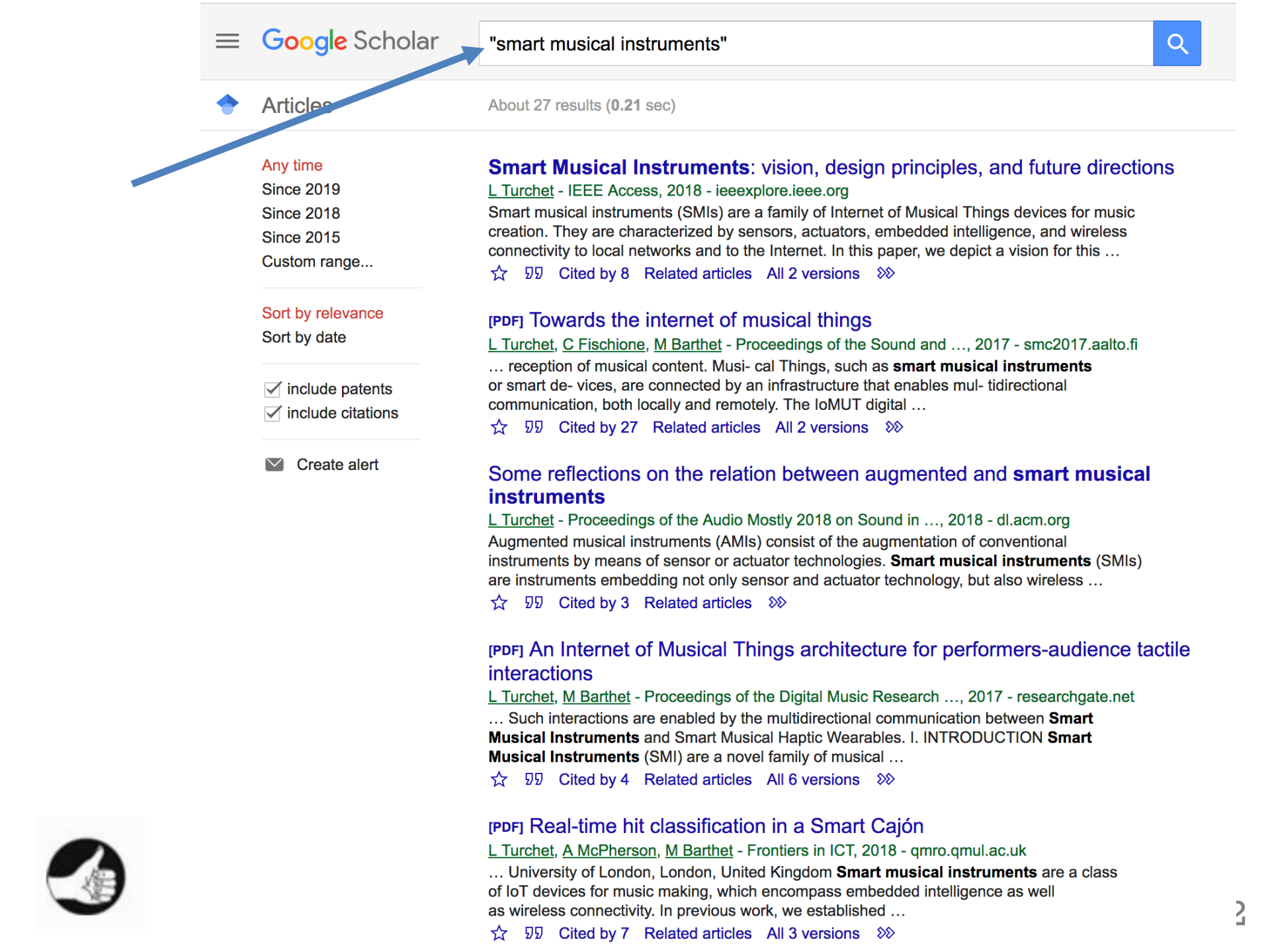


FIGURE 7.3 Slate.com search results: (A) in 2007, users' search terms were not shown, but (B) in 2013, search terms are shown.


Google Scholar
"smart musical instruments"
Articles
About 27 results (0.21 sec)
Any time
Since 2019
Since 2018
Since 2015
Custom range...
Sort by relevance
Sort by date
include patents
include citations
Create alert
Smart Musical Instruments: vision, design principles, and future directions
L Turchet - IEEE Access, 2018 - ieeexplore.ieee.org
Smart musical instruments (SMIs) are a family of Internet of Musical Things devices for music creation. They are characterized by sensors, actuators, embedded intelligence, and wireless connectivity to local networks and to the Internet. In this paper, we depict a vision for this …
Cited by 8 Related articles All 2 versions
[PDF] Towards the internet of musical things
L Turchet, C Fischione, M Barthet - Proceedings of the Sound and …, 2017 - smc2017.aalto.fi
… reception of musical content. Musi- cal Things, such as smart musical instruments or smart de- vices, are connected by an infrastructure that enables mul- tidirectional communication, both locally and remotely. The IoMUT digital …
Cited by 27 Related articles All 2 versions
Some reflections on the relation between augmented and smart musical instruments
L Turchet - Proceedings of the Audio Mostly 2018 on Sound in …, 2018 - dl.acm.org
Augmented musical instruments (AMIs) consist of the augmentation of conventional instruments by means of sensor or actuator technologies. Smart musical instruments (SMIs) are instruments embedding not only sensor and actuator technology, but also wireless …
Cited by 3 Related articles
[PDF] An Internet of Musical Things architecture for performers-audience tactile interactions
L Turchet, M Barthet - Proceedings of the Digital Music Research …, 2017 - researchgate.net
… Such interactions are enabled by the multidirectional communication between Smart Musical Instruments and Smart Musical Haptic Wearables. I. INTRODUCTION Smart Musical Instruments (SMI) are a novel family of musical …
Cited by 4 Related articles All 6 versions
[PDF] Real-time hit classification in a Smart Cajón
L Turchet, A McPherson, M Barthet - Frontiers in ICT, 2018 - qmro.qmul.ac.uk
… University of London, London, United Kingdom Smart musical instruments are a class of IoT devices for music making, which encompass embedded intelligence as well as wireless connectivity. In previous work, we established …
Cited by 7 Related articles All 3 versions

Calls to action (CTAs)

- A UI may display messages to ask users to do something
- CTAs are crucial for guiding users to take desired actions on web pages or applications
- They are interactive elements that prompt users to perform specific tasks, such as making a purchase, signing up for a newsletter, or contacting support
- Effective CTAs are designed to be easily noticeable and compelling, encouraging users to click and move forward with the intended goal
- Competing CTAs (e.g. buttons prompting users to take specific actions) can be detrimental to user experience
- Excessive or conflicting CTAs can overwhelm users, making it difficult for them to choose and take the desired action

Calls to action: guidelines

- **Restrict each message to one action**
- If a message contains multiple requests the user may focus on one of them (usually the first one), get absorbed in responding to that request, and forget to respond to the rest of the requests
  - This is the direct result of the limited capacity of human attention
- **Avoid putting competing calls to action on a page**
  - Each page should have only one dominant call to action – or one for each possible user goals – to not overwhelm users' attention capacity and cause them go down paths that don't achieve their (or the site owner's) goals
- A related guideline: once users have specified their goal, **don't distract them from accomplishing it** by displaying extraneous links and calls to action; instead, **guide them to the goal**

Instructions

- Interactive systems that display instructions for multistep operations should **allow people to refer to the instructions while executing them until completing all the steps**



FIGURE 7.5 Instructions for Windows XP wireless setup start by telling users to close the instructions.

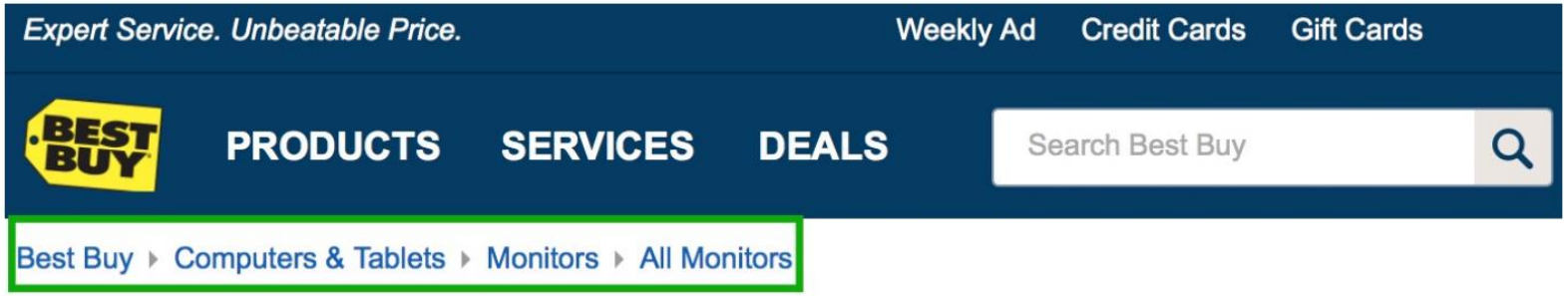


FIGURE 7.4 Instructions in Windows Help files remain displayed while users follow them.

Navigation depth

- It is well established that navigation hierarchies that are broad and shallow are easier for most people to find their way around in than narrow, deep hierarchies (e.g., menus)
- Guideline: in hierarchies deeper than two levels, provide navigation "breadcrumb" paths to **constantly remind users where they are**





Implications of long-term memory characteristics for UI design

- **Long-term** memory has many weaknesses
  - Error-prone
  - Retroactively alterable
  - Easily biased by a variety of factors at the time of recording or retrieval (e.g., emotions)
- Main implication: **people need tools to augment it**
  - Designers need to provide software that fulfils such need
  - Designers should avoid developing systems that overload long-term memory
  - E.g., authentication is a memory task
    - Biometric systems are a better solution than recalling passwords or answers to security questions

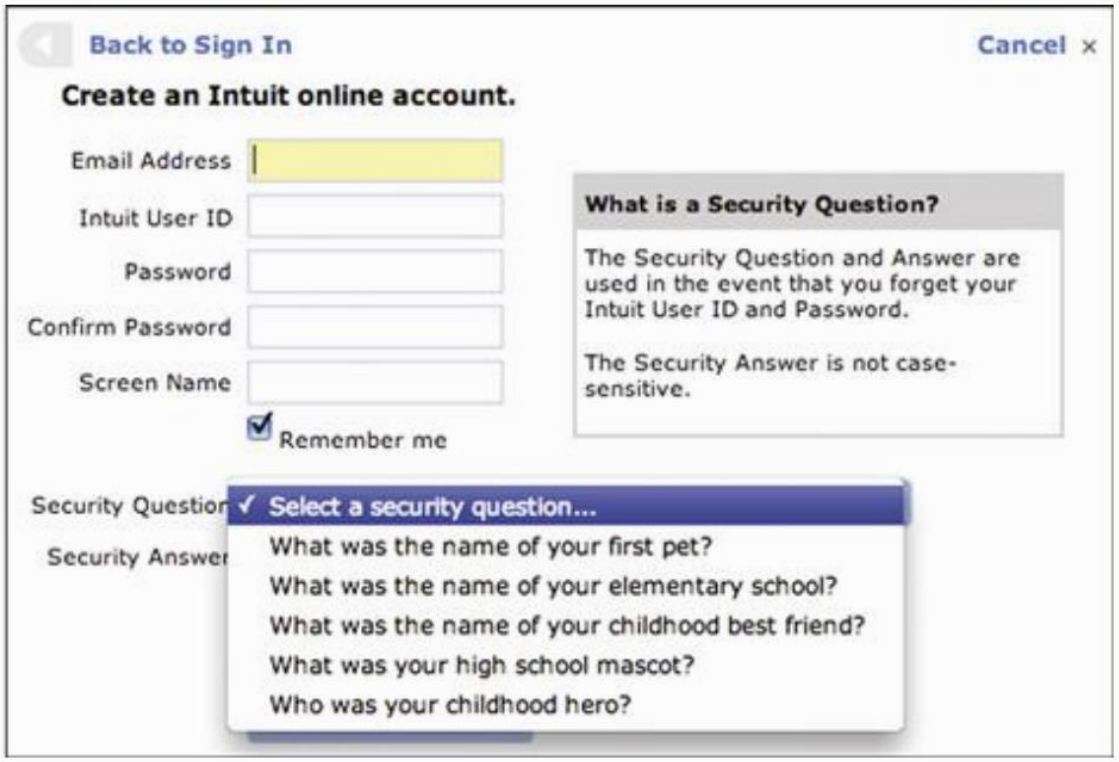


FIGURE 7.7 Intuit.com's registration burdens long-term memory: users may have no unique, memorable answer for any of the questions.

- Some questions may have several possible answers
- **Allow users to create their own security questions**

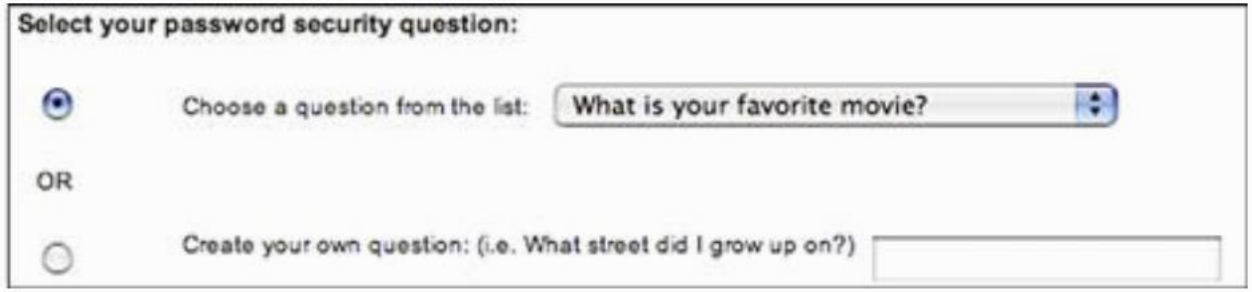


FIGURE 7.8 NetworkSolutions.com lets users create a security question if none on the menu works for them.

Consistency is crucial

- Learning and long-term retention are enhanced by the consistency of the UI
- **The more consistent** the operation of different functions, or the more consistent the actions on different types of objects, **the less users have to learn**
- UIs that have many exceptions and little consistency from one function or object to another require users to store in their long-term memory many features about each function or object and its correct usage context
- The need to encode many features
  - Makes UI harder to learn
  - Makes it more likely that a user's memory will drop essential features during storage or retrieval, increasing the chances that the user will fail to remember, misremember, or make other memory errors

Limits on attention shape our thought and action

- When people interact with computer systems, some aspects of their **behavior** follow predictable **patterns**, some of which result from the limited capacity of attention and short-term memory

- We will see the following:
  - **We focus on our goals and pay little attention to our tools**
  - **We notice things more when they are related to our goals**
  - **We use external aids to keep track of what we are doing**
  - **We follow the *information "scent"* toward our goal**
  - **We prefer familiar paths**
  - **Our thought cycle: goal, execute, evaluate**
  - **After we achieve a task's primary goal, we often forget cleanup steps**

We focus on our goals and pay little attention to our tools

- When people are doing a task – trying to accomplish a goal – most of their attention is focused on the goals and data related to that task
- People think about their tools only superficially, and only when necessary
- When people refocus their attention on their tools, it is pulled away from the details of the task: this shift increases the chances of users losing track of what they were doing or exactly where they were in doing it
- Guideline: software applications and most Web sites should not call attention to themselves; **they should fade into the background and allow users to focus on their own goals**

We notice things more when they are related to our goals

- Perceptual filtering and biasing (because of goals), attention, and memory are closely related
- When an event happens, the few details about it that we notice and remember later are usually those that were important for our goals at the time of the event
- This is demonstrated by two related psychological phenomena
  - **Inattentional blindness**
    - Demonstration: https://youtu.be/vJG698U2Mvo
    - Demonstration: https://youtu.be/IGQmdoK_ZfY
  - **Change blindness**
    - Demonstration: https://www.youtube.com/watch?v=VkrrVozZR2c

Inattentional blindness

- When our mind is intensely occupied with a task, goal, or emotion, we sometimes fail to notice objects and events in our environment that we otherwise would have noticed and remembered



FIGURE 8.1 Scene from video used in "invisible gorilla" study. *Figure provided by Daniel Simons.* For more information about the study and to see the video, go to www.dansimons.com or www.theinvisiblegorilla.com.

Change blindness

- The experiments show that **often, people don't notice differences in features other than those their goals made them pay attention to**

- Implications: when people interact with software, electronic appliances, or online services, it is not uncommon for them to fail to notice important changes in what is displayed
  - E.g., in a experiment involving older adults using travel websites, changes in price resulting from user actions were often not obvious. Even after spotting the price information, participants were prone to change blindness: when they changed trip options (e.g., departure city, additional excursions, cabin class), they often would not notice that the price changed

Change blindness

- Guideline: **make changes obvious** – that is, highly salient – and **take steps to draw users' attention to the change**
  - E.g., a way to draw users' attention to a new error message is to vibrate it briefly when it first appears or highlight it briefly before it reverts to a "normal" appearance

We use external aids to keep track of what we are doing

- E.g., arithmetic: we make use of calculators rather than relying on the limits of our memory and attention to make some calculations
- Implication: **the UI should indicate what users have done versus what they have not yet done**
  - E.g., e-mail applications mark differently read and unread messages
  - E.g., websites mark differently visited and unvisited links
- Implication: **the UI should allow users to mark or move objects** to indicate which ones they have worked on versus which ones they have not worked on

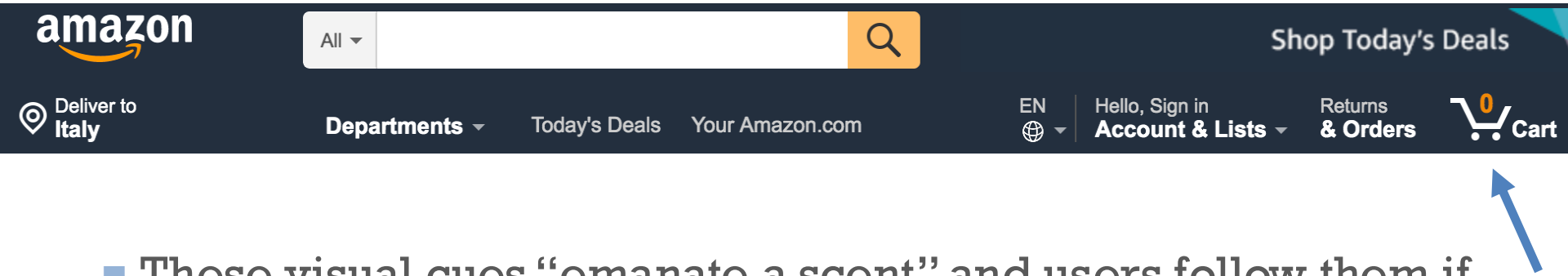
We follow the information "scent" toward our goal

- Our perception is skewed to detect information related to our goals
- **Information scent** refers to the strength of a cue or series of cues which lead to a point within a UI
- A strong information scent is found when an element within the design **suggests the users that they are moving nearer to their goal**



We follow the information "scent" toward our goal

- Example
  - If an e-Commerce site user wants to save an item to their basket they will look for something in the display that explicitly says "save to basket" or a command button with an image of a basket/trolley/cart



  - These visual cues "emanate a scent" and users follow them if they are associated with what they are trying to do in the website

We follow the information "scent" toward our goal

- Interactive systems should be designed **so that the scent is strong and really leads users to their goals**
- To do that, designers need to

1. **understand the goals** that users are likely to have at each decision point in a task
2. **ensure** that each choice point in the software provides options for every important user goal and clearly indicates which option leads to which goal

We follow the information "scent" toward our goal

- Example
  - When a user is navigating an e-Commerce site they will need **ready access** to their shopping basket **at all times**, so they can determine how many items they have, which items they have and how much they will cost individually and in total
  - Also when they are reading product information the "Buy" button should be **prominent**, so the user **does not have to invest any extra time or effort** before they can make a purchase

We follow the information "scent" toward our goal

- There are lots of different ways to **lead the user along their desired goal path**
- …but there are also many ways to **crush their confidence** in the information scent
- To ensure the user does not lose confidence or patience, it is important to provide informative links that **are clearly visible at the appropriate stage of a task**

We prefer familiar paths

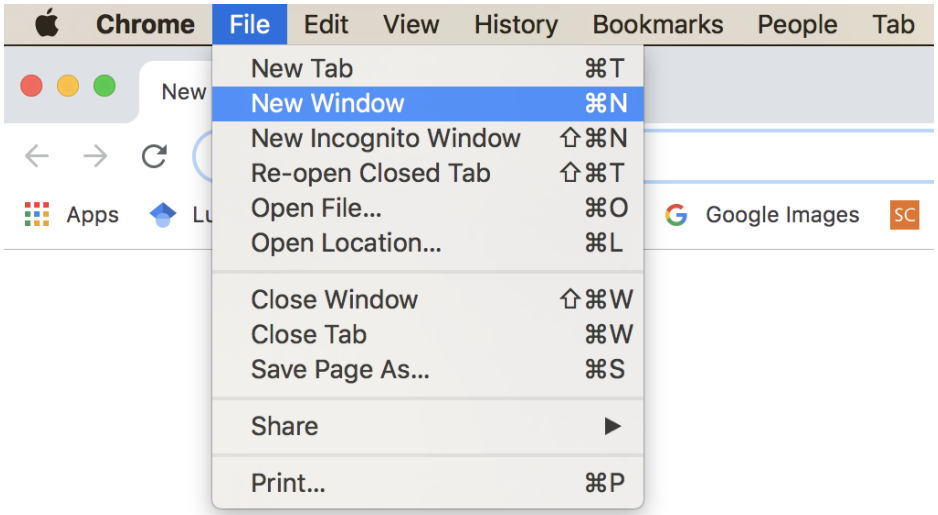
- While pursuing a goal, **people take familiar paths whenever possible rather than exploring new ones**, especially when working under deadlines
- Exploring new paths requires problem solving, which severely taxes our attention and short-term memory
- In contrast, taking familiar, well-learned routes can be done fairly automatically and does not consume much attention and short-term memory

We prefer familiar paths

- Once we learn one way to perform a certain task using a software application, we may continue to do it that way and never discover a more efficient way
- Even if we discover or are told that there is a "better" way, we may stick with the old way because it is familiar, comfortable, and, most importantly, requires little thought
- **Avoiding thought when using computers is important**

We prefer familiar paths: implications

- **Guide users to the best paths**
  - From its first screen or homepage, software should show users the way to their goals; this is basically the guideline that software should provide a clear information scent
- **Help experienced users speed up**
  - Make it easy for users to switch to faster paths after they have gained experience
  - This is why most applications show the keyboard accelerators (e.g., shortcuts) for frequently used functions in the menu-bar menus



Our thought cycle: goal, execute, evaluate

- Scientists studying human behavior have found a cyclical pattern that seems to hold across a wide variety of activities
  1. **Form a goal**
  2. **Execute** actions to try to make progress toward the goal
  3. **Evaluate** whether the actions worked – that is, whether the goal has been reached or is nearer than before
  4. **Repeat** until the goal is reached (or appears unreachable)

Support users in carrying out the goal – execute – evaluate cycle

1. **Goal**. Provide clear paths – including initial steps – for the user goals that the software is intended to support

2. **Execute**. Software concepts (objects and actions) should be based on the task rather than the implementation
   - Don't force users to figure out how the software's objects and actions map to those of the task
   - Provide a clear information scent at choice points to guide users to their goals
   - Don't make them choose actions that seem to take them away from their goal to achieve it

Support users in carrying out the goal – execute – evaluate cycle

3. **Evaluate**. Provide feedback and status information to show users their progress toward the goal; allow users to back out of tasks that didn't take them toward their goal



FIGURE 8.7 ITN.com's flight reservation system clearly shows users' progress toward making a reservation.

After we achieve a task's primary goal, we often forget cleanup steps

- Attention is a very scarce resource; our brain does not waste it by keeping it focused on anything that is no longer important
- When we complete a task, the attentional resources focused on that task's main goal are freed to be refocused on other things that are now more important
- The impression we get is that once we achieve a goal, everything related to it often immediately "falls out" of our short-term memory – that is, we forget about it

After we achieve a task's primary goal, we often forget cleanup steps

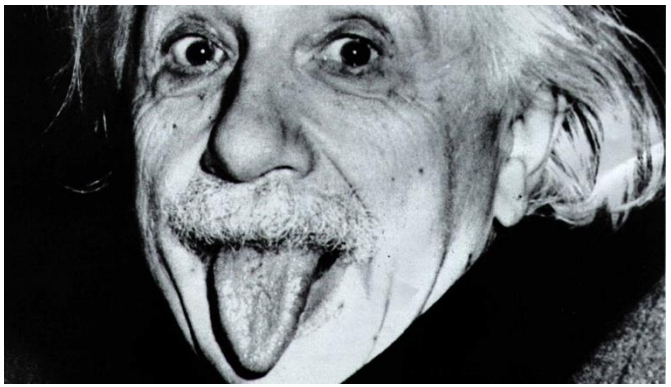
- One result is that people often forget loose-ends of tasks
  - E.g., turn car headlights off after arrival
- These end-of-task short-term memory lapses are **completely predictable and avoidable**
- To avoid such lapses, interactive systems can and should be designed to **remind people that loose-end steps remain**

After we achieve a task's primary goal, we often forget cleanup steps

- In some cases, it may even be possible for the **system to complete the task itself**
  - E.g., cars should turn off headlights automatically when the car is no longer in use, or at least remind drivers that the lights are still on
- Software designers should **consider** whether the tasks supported by a system they are designing have cleanup steps that users are likely to forget, and if so, they should design the system either to help users remember, or eliminate the need for users to remember

Recognition is easy, recall is hard

- Patterns of neural activity, which is what memories are, can be activated in two different ways
  - by more perceptions coming in from the senses: **recognition**
  - by other brain activity: **recall**
- Recognition is easy for the human brain
  - E.g., humans can recognize faces very quickly



  - Recall is hard
    - Because people are bad at recall, they develop methods and technologies to help them remember facts and procedures

GUIs are based on 2 UI design guidelines

1) See and choose is easier than recall and type

- Show users their options and let them choose among them, rather than force users to recall their options and tell the computer what they want
- This rule is the reason GUIs have almost replaced command-line user interfaces (CLIs) in personal computers

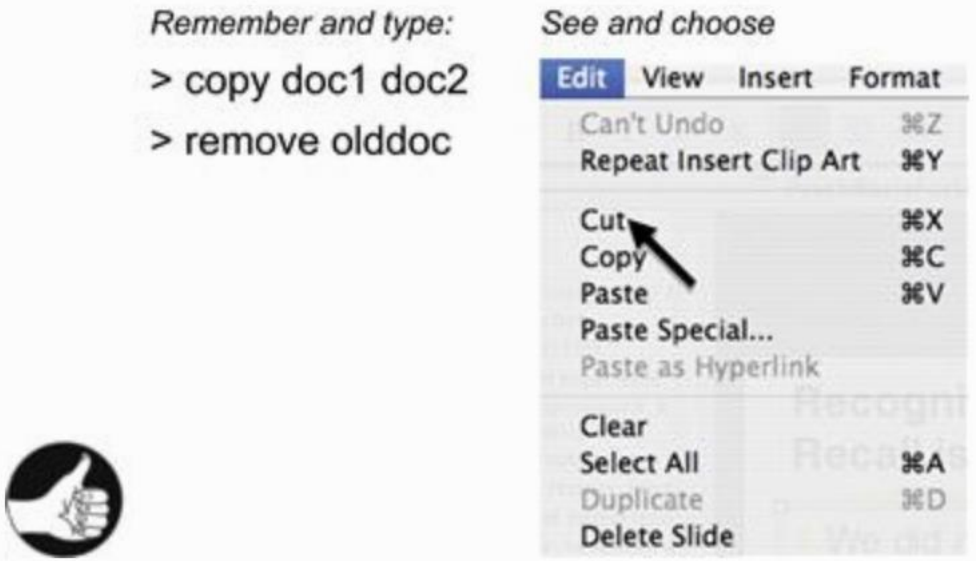


FIGURE 9.5 The main design rule behind today's GUI: "See and choose is easier than remember and type."

GUIs are based on 2 UI design guidelines

2) Use pictures where possible to convey function

- People recognize pictures very quickly, which also stimulates the recall of associated information; for this reason, today's user interfaces often use pictures to convey function



FIGURE 9.6 Desktop icons convey function via recognition—by analogy with physical objects or by experience.



FIGURE 9.7 Wordpress.com uses symbols plus text to label functional pages on the Dashboard.

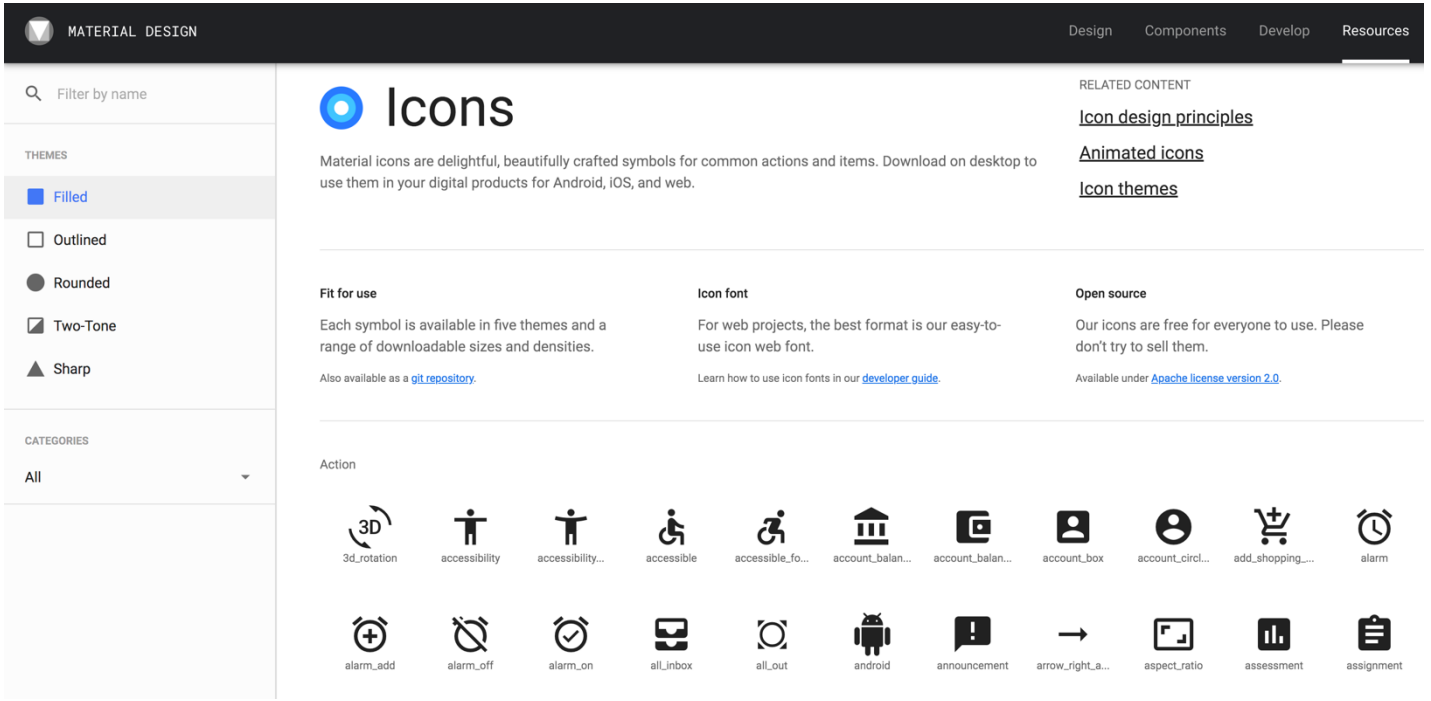
Lesson for the course "Human-Computer Interaction"

Google Material Design (Part 2)

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Icons

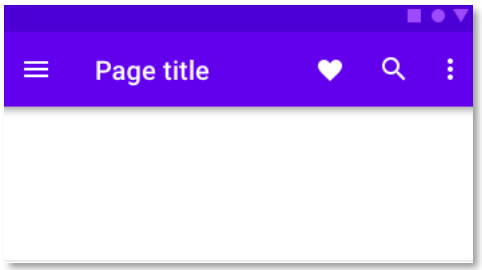


- https://material.io/resources/icons/?style=baseline
- You can download them in SVG format, but you can't change their color directly in Axure (instead, use an opens source software such as Gimp: https://www.gimp.org/ or Inksape: https://inkscape.org/)

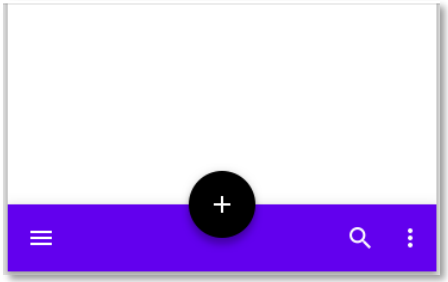
Components

- Material Components are interactive building blocks for creating a user interface
- https://material.io/components/
- Examples:

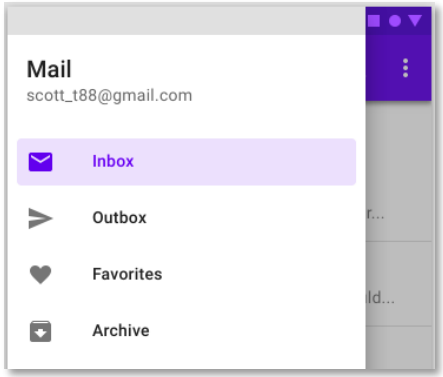
app bar: top



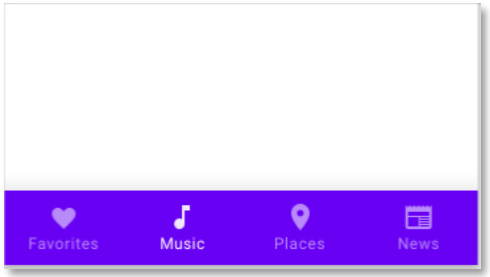
app bar: bottom



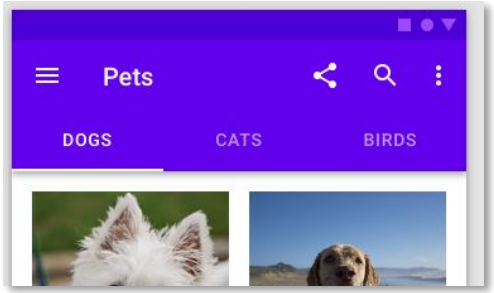
navigation drawer



bottom navigation



tabs



Components in Axure RP

- **Install the Google Material Design Components for Axure RP 9:**
  - https://forum.axure.com/t/material-design-widget-library-for-axure-9/56296
  - Follow the instructions there in (download for Axure RP 9)
  - If you don't have the Roboto font already installed on your machine, then google how to install it
  - The repository is available at: https://github.com/duzyn/material-axure-library

App bars: top

- https://material.io/components/app-bars-top/

Tabs

- https://material.io/components/tabs/

App bars: bottom

- https://material.io/components/app-bars-bottom/

Navigation drawer

- https://material.io/components/navigation-drawer/

Dialogs

- https://material.io/components/dialogs/

Homework

- Read the documentation pages of Material Design seen in this lesson:
  - https://material.io/components/app-bars-top/
  - https://material.io/components/tabs/
  - https://material.io/components/app-bars-bottom/
  - https://material.io/components/navigation-drawer/
  - https://material.io/components/dialogs/

Homework

- Select a software and assess whether the UI help people remember essential information from one moment to the next
  - For instance, does the UI require people to remember system status (or what they have done) or it provides help to remember it?
- Select a software or an electronics appliance (e.g., a microwave) and assess whether there are modes and how they are handled
  - Does the UI provide adequate feedback about which mode the system is in?

Homework

- Select a website and assess whether it provides navigation "breadcrumb" paths to constantly remind users where they are
- Select a website and assess the information scent towards a goal that you want to perform on it (e.g., purchasing an item)
- Select an interactive system (e.g., website) and assess whether there are potential loose-end steps that remain, for which the system does not provide a reminder to complete or does not complete the task itself

Bibliography

- Chapters 7, 8, and 9 of the book *"Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines"* - 2nd Edition - by Jeff Johnson

- https://www.interaction-design.org/literature/article/web-user-behaviour-directed-by-information-scent